Definitions, Theorems, Proofs, and Proof Writing

Math Club

September 22, 2026

Motivation

Question: Is the sum of two odd numbers even or odd?

Examples:

$$2+3=4 \qquad 7+5=13 \qquad 5+1=6 \qquad -3+1=-2$$

With further examples we see that the pattern holds

How can we prove this is true for *all* integers without testing each case?

Theorems

First, we must explicitly write what we are trying to prove. A **theorem** is a mathematical statement that has been proven to be true on the basis of previously established statements. **Theorem:**

The sum of two odd numbers is even. However, for rigor and

completeness we must know exactly what an even or odd number is.

Definitions

A theorem must be based in precise and agreed upoon definitions. A **definition** establishes the exact, non-negotiable meaning of a term. It is either satifised or not.

Definitions are **agreed upon** as a fact rather than proven. Our theorem and proofs are then built upon these.

Definitions: Parity of Integers

Recall the set of integers is $\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$.

Definition: Even and Odd Integers
Let $n \in \mathbb{Z}$.

- n is **even** if $n = 2k$ for some integer $k \in \mathbb{Z}$.
- n is **odd** if $n = 2k + 1$ for some integer $k \in \mathbb{Z}$.

This definition:

- Sets the boundary to apply only to integers
- Paritions all integers into being either even or odd with no in between
- Provides us with variables and mathematical expressions that can be manipulated

The Logic of a Direct Proof

Most mathematical claims can be structured as conditional statements:

$$P \implies Q \quad \text{“If P, then Q”}$$

- **Hypothesis (P):** What you are given as true (e.g., *“n and m are odd integers”*).
- **Conclusion (Q):** What you want to deduce (e.g., *“$n + m$ is an even integer”*).

The Logical Chain: A direct proof bridges P to Q through a sequence of valid deductions:

$$P \implies A_1 \implies A_2 \implies \cdots \implies A_n \implies Q$$

P is true by assumption, so if our logic is correct in each step, all following statements are true, thus leaving us with Q

Scratchwork

Before writing a formal proof, test your ideas and map out the logic on scratch paper. **Workflow**

- **Unpack Definitions:** Translate the assumptions (P) using definitions
- **Identify Target State:** Detirmine what the conclusion (Q) needs to look like structurally.
- **Manipulate & Connect:** Apply algebra, logic, etc to reach Q.